

- RAG Learning

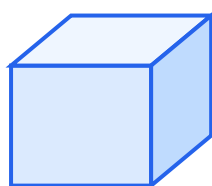
4.4 Mensuration: Area, Volume & 3D

Pearson Edexcel GCSE Mathematics 1MA1 — Geometry and measures — spec refs G12–G14, G16, G17 — Papers 1 (non-calculator), 2 and 3 (calculator)

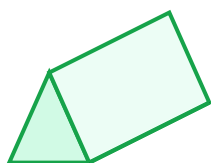
Mensuration means measuring: lengths, areas and volumes. In the Edexcel 1MA1 specification the statements G12–G14, G16 and G17 are **identical at Foundation and Higher**, so every formula on this page is examinable on both tiers — including cones, spheres, pyramids and composite solids. What makes a Higher question harder is the extra content it is combined with: Pythagoras in 3D, area and volume scale factors for similar solids, bounds, surds and algebra. Those links are collected in the final section, which is the only part tagged Higher.

4.4.1 Faces, Edges and Vertices of 3D Solids (G12)

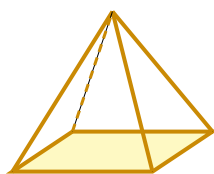
Face — a flat surface of a solid. **Surface** — any outside surface, flat or curved. **Edge** — a line where two faces or surfaces meet. **Vertex** (plural vertices) — a point where edges meet.



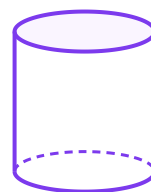
Cuboid



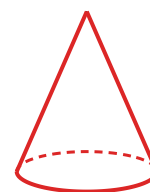
Triangular prism



Square-based pyramid



Cylinder



Cone

Solid	Faces / surfaces	Edges	Vertices
Cube / cuboid	6 faces	12	8
Triangular prism	5 faces (2 triangles, 3 rectangles)	9	6

Pentagonal prism	7 faces (2 pentagons, 5 rectangles)	15	10
Square-based pyramid	5 faces (1 square, 4 triangles)	8	5
Tetrahedron (triangular-based pyramid)	4 faces	6	4
Cylinder	2 flat circles + 1 curved surface	2 (curved)	0
Cone	1 flat circle + 1 curved surface	1 (curved)	1 (apex)
Sphere	1 curved surface	0	0

A **prism** has the same cross-section all the way along its length. A prism whose cross-section has n sides has $n + 2$ faces, $3n$ edges and $2n$ vertices. A **pyramid** on an n -sided base has $n + 1$ faces, $2n$ edges and $n + 1$ vertices. For any solid with flat faces, Euler's formula $F + V - E = 2$ is a useful check.

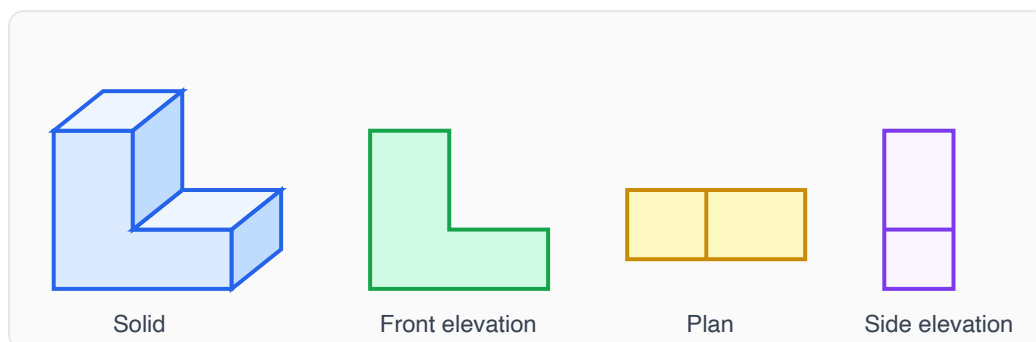
Worked Example: A prism has a hexagonal cross-section. How many faces, edges and vertices does it have? Check with Euler's formula.

$$n = 6: \text{faces} = 6 + 2 = \mathbf{8}, \text{edges} = 3 \times 6 = \mathbf{18}, \text{vertices} = 2 \times 6 = \mathbf{12}$$

$$\text{Check: } 8 + 12 - 18 = 2 \checkmark$$

4.4.2 Plans and Elevations (G13)

Plan — the view from directly above. **Front elevation** — the view from the front. **Side elevation** — the view from the side. Each is a 2D drawing that shows true lengths as seen from that direction.



Draw a solid line on a plan or elevation wherever there is a visible edge — a change in height or depth. Edexcel questions ask you both to *construct* these views on a grid and to *interpret* them (for example, use a plan and elevation to work out a volume or sketch the solid).

Worked Example: A triangular prism is 8 cm long and lies on one of its rectangular faces. Its cross-section is an isosceles triangle with base 4 cm and height 3 cm. Describe its plan, front elevation (looking at the 8 cm side) and side elevation.

Plan: a **rectangle 8 cm by 4 cm** with a line along the middle (the top ridge).

Front elevation: a **rectangle 8 cm by 3 cm**.

Side elevation: an **isosceles triangle** with base 4 cm and height 3 cm.

Exam Tip: Lengths must match between views: the width of the plan equals the width of the front elevation, and the depth of the plan equals the width of the side elevation. Use a ruler and count squares carefully on isometric or squared paper.

4.4.3 Standard Units of Measure (G14)

Length, area and volume are measured in different units. Because area is two lengths multiplied and volume is three, the length conversion factor must be **squared** for area and **cubed** for volume.

Length	Area	Volume / capacity
1 cm = 10 mm	$1 \text{ cm}^2 = 10^2 = 100 \text{ mm}^2$	$1 \text{ cm}^3 = 10^3 = 1000 \text{ mm}^3$

1 m = 100 cm	1 m ² = 100 ² = 10 000 cm ²	1 m ³ = 100 ³ = 1 000 000 cm ³
1 km = 1000 m	1 km ² = 1 000 000 m ²	1 litre = 1000 cm ³ 1 ml = 1 cm ³
		1 m ³ = 1000 litres

G14 also covers mass (1 kg = 1000 g, 1 tonne = 1000 kg), time (1 hour = 60 minutes = 3600 seconds) and money (£1 = 100p).

Worked Example: (a) Change 2.6 m² into cm². (b) Change 7 500 000 cm³ into m³. (c) A jug holds 4.8 litres. Write this in cm³.

(a) $2.6 \times 10\,000 = \mathbf{26\,000\,cm^2}$

(b) $7\,500\,000 \div 1\,000\,000 = \mathbf{7.5\,m^3}$

(c) $4.8 \times 1000 = \mathbf{4800\,cm^3}$

Exam Tip: Never convert 1 m² to 100 cm². A square 1 m by 1 m is 100 cm by 100 cm, which is 10 000 cm². If in doubt, convert the lengths first and then multiply.

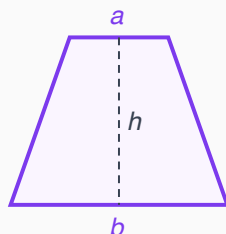
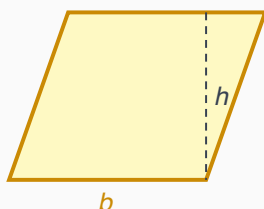
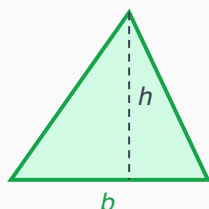
4.4.4 Perimeter and Area of 2D Shapes (G16, G17)

Perimeter — the total distance around the edge of a shape (cm, m). **Area** — the amount of flat space it covers (cm², m²).

Shape	Area (you must know these — G16)
Rectangle	$A = lw$
Triangle	$A = \frac{1}{2}bh$ (h = perpendicular height)
Parallelogram	$A = bh$ (h = perpendicular height)

Trapezium

$$A = \frac{1}{2}(a + b)h \quad (a, b = \text{parallel sides})$$



Worked Example: A trapezium has parallel sides 5 cm and 13 cm. The perpendicular distance between them is 8 cm. Work out its area.

$$A = \frac{1}{2}(5 + 13) \times 8 = \frac{1}{2} \times 18 \times 8 = \mathbf{72 \text{ cm}^2}$$

Circles

You must know (G17): Circumference $C = 2\pi r = \pi d$ Area $A = \pi r^2$

Worked Example: A circular table top has radius 9 cm. Work out its circumference and area, each to 1 decimal place.

$$C = 2 \times \pi \times 9 = 18\pi = 56.548\dots \approx \mathbf{56.5 \text{ cm}}$$

$$A = \pi \times 9^2 = 81\pi = 254.469\dots \approx \mathbf{254.5 \text{ cm}^2}$$

Worked Example: A semicircle has diameter 14 cm. Work out its perimeter and its area, in terms of π .

$$\text{Curved edge} = \frac{1}{2} \times \pi \times 14 = 7\pi, \text{ so perimeter} = \mathbf{7\pi + 14 \text{ cm}} \text{ (include the diameter)}$$

$$\text{Area} = \frac{1}{2} \times \pi \times 7^2 = \mathbf{24.5\pi \text{ cm}^2}$$

"Give your answer **in terms of π** " means leave π as a symbol, e.g. $24.5\pi \text{ cm}^2$. This is an exact answer and is common on Paper 1.

Composite Shapes

Split the shape into rectangles, triangles, trapezia and parts of circles. Add the areas of parts that are joined on; subtract the areas of parts that are cut out. For perimeter, only add the edges on the outside.

Worked Example: A rectangle 16 cm by 10 cm has a semicircle of diameter 10 cm cut out of one of its short sides. Work out the area that is left, to 1 decimal place.

$$\text{Rectangle} = 16 \times 10 = 160 \text{ cm}^2$$

$$\text{Semicircle} = \frac{1}{2} \times \pi \times 5^2 = 12.5\pi = 39.26 \dots \text{ cm}^2$$

$$\text{Area left} = 160 - 12.5\pi = 120.73 \dots \approx \mathbf{120.7 \text{ cm}^2}$$

Areas on Coordinate Axes

Horizontal lengths are differences in x -coordinates and vertical lengths are differences in y -coordinates.

Worked Example: A triangle has vertices $(-1, 2)$, $(5, 2)$ and $(3, 8)$. Work out its area.

The base from $(-1, 2)$ to $(5, 2)$ is horizontal: $5 - (-1) = 6$. The perpendicular height is $8 - 2 = 6$.

$$\text{Area} = \frac{1}{2} \times 6 \times 6 = \mathbf{18 \text{ units}^2}$$

4.4.5 Volume of Cuboids, Prisms and Cylinders (G16)

Right prism: $V = \text{area of cross-section} \times \text{length}$

Cuboid: $V = lwh$ **Cylinder:** $V = \pi r^2 h$ (a cylinder is a prism with a circular cross-section)

Worked Example: A prism is 12 cm long. Its cross-section is a trapezium with parallel sides 4 cm and 10 cm, 5 cm apart. Work out its volume.

$$\text{Cross-section} = \frac{1}{2}(4 + 10) \times 5 = 35 \text{ cm}^2; \quad V = 35 \times 12 = \mathbf{420 \text{ cm}^3}$$

Worked Example: A tin is a cylinder with diameter 8 cm and height 15 cm. Work out its volume to 3 significant figures.

$$r = 4; \quad V = \pi \times 4^2 \times 15 = 240\pi = 753.98 \dots \approx \mathbf{754 \text{ cm}^3}$$

Worked Example: A cuboid tank 120 cm by 60 cm by 50 cm contains water 35 cm deep. How many more litres are needed to fill it?

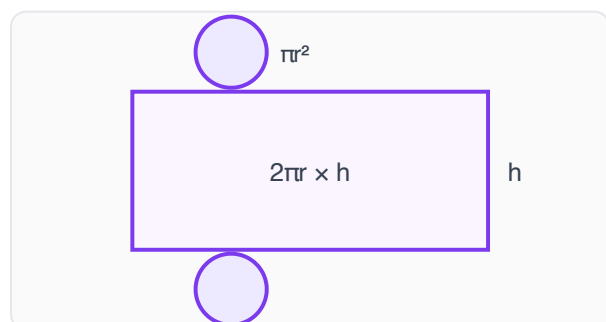
$$\text{Unfilled depth} = 50 - 35 = 15 \text{ cm, so extra volume} = 120 \times 60 \times 15 = 108\,000 \text{ cm}^3 = \mathbf{108 \text{ litres}}$$

4.4.6 Surface Area of Prisms and Cylinders (G16, G17)

Surface area is the total area of all the outside faces. Sketching the **net** makes sure no face is missed.

Cuboid: $SA = 2(lw + lh + wh)$

Cylinder: curved surface = $2\pi rh$; closed cylinder total = $2\pi r^2 + 2\pi rh$



Worked Example: A closed cylinder has radius 3 cm and height 10 cm. Work out its total surface area in terms of π .

Two ends: $2 \times \pi \times 3^2 = 18\pi$. Curved surface: $2 \times \pi \times 3 \times 10 = 60\pi$.

Total = **$78\pi \text{ cm}^2$**

Worked Example: A prism 15 cm long has a cross-section that is a right-angled triangle with sides 6 cm, 8 cm and 10 cm. Work out its surface area.

Two triangles: $2 \times \frac{1}{2} \times 6 \times 8 = 48$. Three rectangles: $(6 + 8 + 10) \times 15 = 360$.

Total = **408 cm^2**

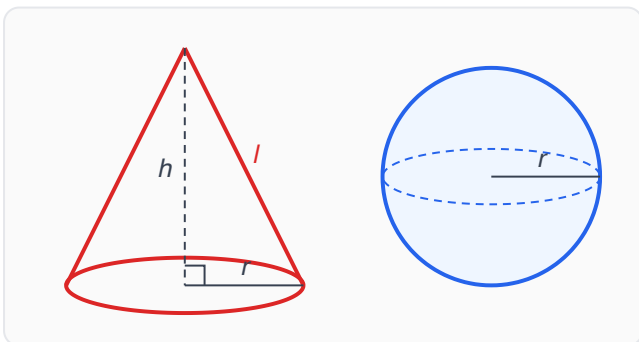
4.4.7 Pyramids, Cones and Spheres (G17)

Pyramid: $V = \frac{1}{3} \times \text{area of base} \times h$

Cone: $V = \frac{1}{3}\pi r^2 h$; curved surface area = $\pi r l$ (l = slant height)

Sphere: $V = \frac{4}{3}\pi r^3$; surface area = $4\pi r^2$

Hemisphere: $V = \frac{2}{3}\pi r^3$; curved surface = $2\pi r^2$



On Edexcel papers the cone and sphere formulae are given in the question when they are needed (they are not on a formulae sheet), but you must be able to use them. The pyramid formula is usually given too. The radius r , perpendicular height h and slant height l of a cone form a right-angled triangle, so $l^2 = r^2 + h^2$.

Worked Example: A square-based pyramid has volume 245 cm^3 and perpendicular height 15 cm. Work out the length of a side of the base.

$$\frac{1}{3} \times x^2 \times 15 = 245 \Rightarrow 5x^2 = 245 \Rightarrow x^2 = 49 \Rightarrow x = \mathbf{7 \text{ cm}}$$

Worked Example: A solid cone has base radius 9 cm and perpendicular height 12 cm. Work out its total surface area in terms of π .

$$\text{Slant height } l = \sqrt{9^2 + 12^2} = \sqrt{225} = 15 \text{ cm}$$

$$\text{Curved surface} = \pi \times 9 \times 15 = 135\pi; \text{ base} = \pi \times 9^2 = 81\pi$$

$$\text{Total} = \mathbf{216\pi \text{ cm}^2}$$

Worked Example: A ball is a sphere of radius 7.5 cm. Work out its volume to 3 significant figures.

$$V = \frac{4}{3} \times \pi \times 7.5^3 = 562.5\pi = 1767.1 \dots \approx \mathbf{1770 \text{ cm}^3}$$

Exam Tip: Volume of a cone uses the **perpendicular** height h ; curved surface area uses the **slant** height l . If the question gives the other one, use Pythagoras first.

4.4.8 Composite Solids and Frustums (G17)

"Composite solids" are named in G17 on **both tiers**. For **volume**, add or subtract the volumes of the parts. For **surface area**, include only the surfaces on the outside — leave out any face where two parts are joined.

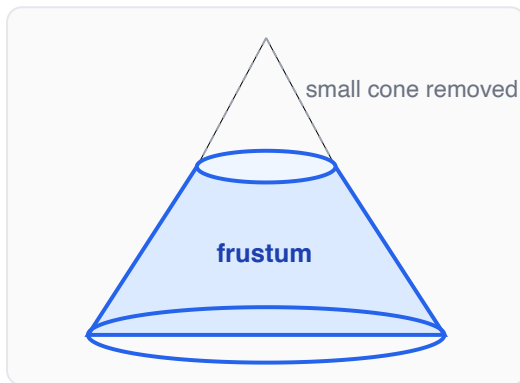
Worked Example: An ice-cream model is a cone of radius 3 cm and perpendicular height 8 cm with a hemisphere of radius 3 cm on top. Work out its volume to 3 significant figures.

$$\text{Cone} = \frac{1}{3} \times \pi \times 3^2 \times 8 = 24\pi; \text{ hemisphere} = \frac{2}{3} \times \pi \times 3^3 = 18\pi$$

$$\text{Total} = 42\pi = 131.9 \dots \approx \mathbf{132 \text{ cm}^3}$$

A **frustum** is the solid left when the top of a cone (or pyramid) is cut off parallel to its base.

$$V_{\text{frustum}} = V_{\text{large cone}} - V_{\text{small cone}}$$



Worked Example: A frustum is made by removing a cone of radius 3 cm and height 4 cm from the top of a cone of radius 12 cm and height 16 cm. Work out its volume to 3 significant figures.

$$\text{Large cone} = \frac{1}{3} \times \pi \times 12^2 \times 16 = 768\pi$$

$$\text{Small cone} = \frac{1}{3} \times \pi \times 3^2 \times 4 = 12\pi$$

$$\text{Frustum} = 768\pi - 12\pi = 756\pi = 2375.0 \dots \approx \mathbf{2380 \text{ cm}^3}$$

If you are given only the two radii and the depth of the frustum, find the height of the whole cone with similar triangles. A lampshade has radii 20 cm and 12 cm and depth 16 cm: $\frac{H - 16}{H} = \frac{12}{20}$ gives $20H - 320 = 12H$, so $H = 40$ cm and the removed cone has height 24 cm. (Frustum questions are most common on Higher papers because they combine G17 with similarity.)

Melting, Recasting and Displacement

Worked Example: A solid metal sphere of radius 9 cm is melted down and recast into cones of radius 3 cm and height 6 cm. How many cones can be made?

$$\text{Sphere} = \frac{4}{3} \times \pi \times 9^3 = 972\pi; \text{cone} = \frac{1}{3} \times \pi \times 3^2 \times 6 = 18\pi$$

$$972\pi \div 18\pi = \mathbf{54 \text{ cones}}$$

Worked Example: A cylindrical jar of radius 5 cm contains water. A metal ball of radius 3 cm is dropped in and is fully covered. By how much does the water level rise?

$$\text{Ball} = \frac{4}{3} \times \pi \times 3^3 = 36\pi \text{ cm}^3. \text{ Rise} = 36\pi \div (\pi \times 5^2) = 36 \div 25 = \mathbf{1.44 \text{ cm}}$$

Exam Tip: Keep π in your working until the end — in "how many" questions it usually cancels. Round *down* for "how many can be made" and *up* for "how many tins are needed".

4.4.9 Mensuration with Higher-Tier Content Higher

These skills come from other Higher-only statements but often appear inside area and volume questions.

Pythagoras in 3D solids (G20)

Worked Example: A box is a cuboid 2 cm by 3 cm by 6 cm. Work out the length of the longest straight rod that fits inside it.

$$\text{Space diagonal} = \sqrt{2^2 + 3^2 + 6^2} = \sqrt{49} = \mathbf{7 \text{ cm}}$$

Area and volume of similar solids (G19)

If lengths are multiplied by scale factor k , areas are multiplied by k^2 and volumes by k^3 .

Worked Example: Two similar jugs have heights 10 cm and 30 cm. The smaller jug has surface area 150 cm² and capacity 400 ml. Work out the surface area and the capacity of the larger jug.

$k = 30 \div 10 = 3$. Area: $150 \times 3^2 = \mathbf{1350 \text{ cm}^2}$. Capacity: $400 \times 3^3 = 10\,800 \text{ ml} = \mathbf{10.8 \text{ litres}}$

Bounds in mensuration (N16)

Worked Example: A cylinder has radius 5.0 cm (to 1 d.p.) and height 12 cm (to the nearest cm). Work out the upper bound of its volume.

Upper bounds: $r = 5.05$, $h = 12.5$. $V = \pi \times 5.05^2 \times 12.5 = 1001.4 \dots \approx \mathbf{1001 \text{ cm}^3}$

Exact answers with surds (N8)

Worked Example: A circle has area $18\pi \text{ cm}^2$. Find its exact radius.

$$\pi r^2 = 18\pi \Rightarrow r^2 = 18 \Rightarrow r = \sqrt{18} = \mathbf{3\sqrt{2} \text{ cm}}$$

Summary of Formulae

Shape	Volume	Surface area
Cuboid	lwh	$2(lw + lh + wh)$
Prism	cross-section $\times$ length	sum of all faces
Cylinder	$\pi r^2 h$	$2\pi r^2 + 2\pi rh$
Pyramid	$\frac{1}{3} \times \text{base} \times h$	base + triangular faces
Cone	$\frac{1}{3}\pi r^2 h$	$\pi rl + \pi r^2$

Sphere	$\frac{4}{3}\pi r^3$	$4\pi r^2$
Hemisphere (solid)	$\frac{2}{3}\pi r^3$	$3\pi r^2$